

CY 511: Group Theory and Spectroscopy

Course Code	CY 511
Course Name	Group Theory and Spectroscopy
L-T-P-C	3-0-0-3
Prerequisites	B.Sc. (with Chemistry) or Teachers Consent
Intended for	UG/PG
Distribution	Core
Approval	8th Senate (Oct 2015)

Course Contents

- **Molecular symmetry and group theory**

Definition of a group and basic theorems, Group multiplication table, elements of a symmetry group, symmetry group classification. The Great Orthogonality theorem Direct products, similarity transformation, Characters of representation, Character table, irreducible representation, symmetry adopted linear combination, bonding in diatomics, group theory and molecular electronic states. [16 Lectures]

- **Interaction of light with matter**

Transition moments and transition probabilities, Einsteins coefficients, oscillator strength, Beer- Lambert law, polarizabilities, oscillator strength, relationship between Einsteins coefficients and total absorbance. orr-Oppenheimer approximation, Frank Condon Principles, [4 Lectures]

- **Rotational and Vibrational Spectroscopy**

Moment of inertia, The Rotational energy levels, Rotational spectra of diatomic molecules, Rotational Raman spectra, Rotational spectra of polyatomic molecules, Molecular vibrations, Selection rules, vibrational spectra of diatomic molecules, simple harmonic oscillator and rigid rotor model, anharmonic corrections, Vibrational-Rotational Spectra, vibrational Raman spectra of diatomic molecules. Infrared absorption spectra of polyatomic molecules, symmetric and asymmetric top molecules, normal modes of vibration and their classification by group theory, coupling between rotational and vibrational degrees of freedom. Symmetry and normal modes of vibrations. Symmetry and selection rules for allowed transitions among rotational, vibrational level. Determining the symmetry of molecular motions. Group theory and molecular electronic transitions. Symmetry and selection rules, spin and parity forbidden transitions, vibronic interaction. [16 Lectures]

- **Electronic spectroscopy**

Electronic angular momentum and magnetic moment, atomic spectroscopy, spectra of hydrogen atom, many electron atoms, spectra and structure of diatomic molecule, electronic spectra of polyatomic molecules. [6 Lectures]

Text Books

1. Donald A. McQuarrie and John D. Simon, **Physical Chemistry: A Molecular Approach**, Viva Books, 1998.
2. Peter Atkins and Julio de Paula, **Physical Chemistry**, 7th Edition, Oxford University Press, 2002.
3. J. M. Hollas, **Modern Spectroscopy**, 4th Edition, John Wiley, 2004.
4. F. A. Cotton, **Chemical Applications of Group Theory**, 3rd Edition, John Wiley, 2003.

References

1. Ira N. Levine, **Quantum Chemistry**, Prentice-Hall: Englewood Cliffs, NJ, USA (1991).
2. Alan Vincent, **Molecular Symmetry and Group Theory**, 2nd Edition, Wiley, 2000.
3. Robert L. Carter, **Molecular Symmetry and Group Theory**, John Wiley & Sons, 1997.
4. K. Verra Reddy, **Symmetry and Spectroscopy of Molecules**, 2nd Edition, New Age International Limited Publishers, 2009.